The documentation and process conversion measures necessary to comply with this revision shall be completed by 4 Apr 94.

INCH-POUND

MIL-S-19500/445C
4 January 1994
SUPERSEDING
MIL-S-19500/445B
24 November 1992

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON, SWITCHING
TYPES 1N5712, 1N5712-1, AND 1N5712UR-1
JAN, JANTX, JANTXV, JANS, JANHC AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers the detail requirements for Schottky barrier diodes intended for use in applications requiring switching speeds of less than 100 pico seconds. Four levels of product assurance are provided for each encapsulated device type as specified in MIL-S-19500. Two level of product assurance are provided for each unencapsulated device type.

1.2 Physical dimensions. See figure 1, similar to DO-35 (1N5712 and 1N5712-1), figure 2 (DO-213AA), and figure 3 (JANHC and JANKC).

1.3 Maximum ratings.

| TYPE | $R_{\theta JL}$ 1/ | $V_{RWM}$ | $I_O$ | $T_{STG}$ and $T_J$ |
|---|---|---|---|---|
| | °C/W | V(pk) | mA dc | |
| 1N5712 | 700 | 16 | 33 2/ | -65°C to +200°C |
| 1N5712-1 | 250 | 16 | 33 3/ | -65°C to +150°C |
| 1N5712UR-1 | 100 | 16 | 33 4/ | -65°C to +150°C |

1/ Lead length = .375 inch except for surface mount.
2/ At $T_L$ = +140°C and L = .375 inch, derate $I_O$ to 0 at +200°C.
3/ At $T_L$ = +130°C and L = .375 inch, derate $I_O$ to 0 at +150°C.
4/ At $T_{EC}$ = +140°C, derate $I_O$ to 0 at +150°C.

1.4 Primary electrical characteristics.

| Limits | $V_{(BR)1}$ $I_R$ = 10 µA dc | $V_{F1}$ $I_F$ = 1 mA dc | $V_{F2}$ $I_F$ = 35 mA dc | $I_{R1}$ $V_R$ = 16 V dc | C $V_R$ = 0, f = 1 MHz $V_{sig}$ = 50 mV(pk) | | $\tau_{CL}$ $I_F$ = 20.0 mA | $Z_{\theta JX}$ | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | pF | | | °C/W | |
| | V dc | V dc | V dc | nA dc | 1N5712 | 1N5712-1 1N5712UR-1 | ps | 1N5712 | 1N5712-1 1N5712UR-1 |
| Minimum | 20 | | | | | | | | |
| Maximum | | 0.55 | 1.0 | 150 | 1.2 | 2.0 | 100 | 100 | 40 |

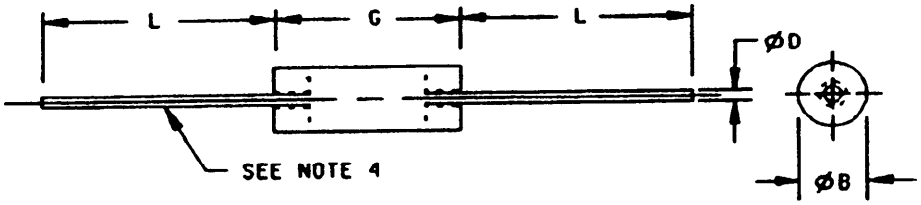
Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: NASA/Parts Projects Office (NPPO), NASA Goddard Space Flight Center, Code 310.A, Greenbelt, MD 20771 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5961

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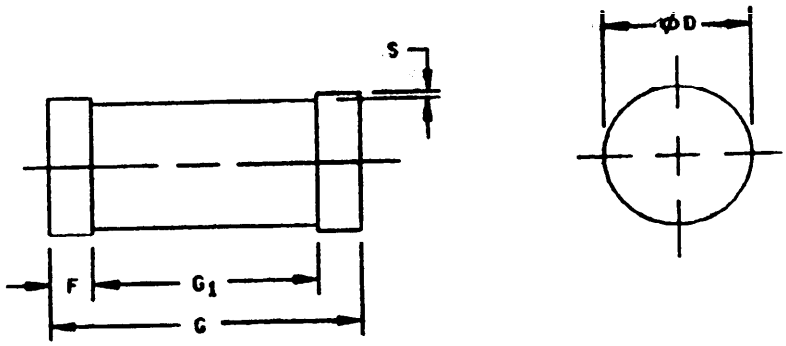

| Dimensions | | | | | |
|---|---|---|---|---|---|
| Ltr | Inches | | Millimeters | | Notes |
| | Min | Max | Min | Max | |
| G | .150 | .170 | 3.81 | 4.32 | 3 |
| ØB | .068 | .076 | 1.73 | 1.93 | 3, 4 |
| L | 1.000 | 1.500 | 25.40 | 38.10 | |
| Ød | .014 | .022 | 0.36 | 0.56 | |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimensions G and ØB include all components of the diode periphery except the sections of the leads over which the diameter is controlled.
4. Dimension ØB shall be measured at the largest diameter.

FIGURE 1. Physical dimensions for 1N5712 and 1N5712-1.



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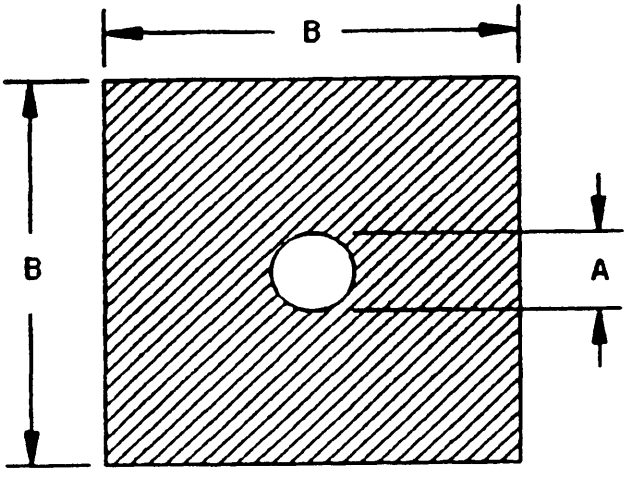



| Dimensions | | | | |
|---|---|---|---|---|
| Ltr | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| ϕD | .063 | .067 | 1.60 | 1.70 |
| F | .016 | .022 | 0.41 | 0.55 |
| G | .130 | .146 | 3.30 | 3.71 |
| G1 | .100 Ref | | 2.54 Ref | |
| S | .001 Min | | 0.03 Min | |

FIGURE 2. Physical dimensions, 1N5712UR-1 (DO-213AA).



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BACKSIDE IS CATHODE

| Dimensions | | | | |
|---|---|---|---|---|
| Ltr | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| A | .0029 | .0035 | .074 | .089 |
| B | .0130 | .0170 | .330 | .430 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Element evaluation accomplished utilizing TO-5 package.
4. The physical characteristics of the die are:

Metallization:
Top (anode): Al
Back (cathode): Au
Al thickness: 25000 Å min
Gold thickness: 4000 Å min
Chip thickness: .010 inch ±.002 inch

FIGURE 3. Physical dimensions, JANHC and JANKC die.



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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWING

Defense Electronics Supply Center

C68001 - Test Fixture for Effective Minority Carrier Lifetime.

(Copies of DESC drawing C68001 are available from Defense Electronics Supply Center, DESC-ECT, Dayton, OH 45444-5270.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500, and as follows:

EC - - - - - - - - - - - End cap.
UR - - - - - - - - - - - Surface mount case outline, round end cap.
$\tau_{CL}$ - - - - - - - - - - - Effective carrier lifetime.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, and figures 1, 2, and 3 herein.

3.3.1 Lead finish. Lead finish shall be solderable as defined in MIL-S-19500, MIL-STD-750, and herein.

3.3.2 Dash one construction. All devices shall be metallurgically bonded double plug construction in accordance with the requirements of category I or II, (see MIL-S-19500).

3.4 Marking. Marking shall be in accordance with MIL-S-19500.



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3.4.1 UR devices. "UR" devices shall be marked with a cathode band only. Initial container package marking will be in accordance with MIL-S-19500.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

4.2.1 Group E inspection. Qualification inspection shall be in accordance with MIL-S-19500 and table II herein.

4.2.2 JANHC and JANKC devices. Qualification shall be in accordance with appendix H of MIL-S-19500.

4.3 Screening (JAN, JANTX, JANTXV, and JANS levels only). Screening shall be in accordance with MIL-S-19500 (table II), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| Screen (see table II of MIL-S-19500) | Measurement | | |
|---|---|---|---|
| | JANS | JANTX and JANTXV | JAN level |
| 3a | Temperature cycling | Temperature cycling | Temperature cycling (per JANTX level) |
| 3c 1/ | Thermal impedance (see 4.3.3) | Thermal impedance (see 4.3.3) | Thermal impedance (see 4.3.3) |
| 9 | Not applicable | Not applicable | |
| 10 | Not applicable | Not applicable | |
| 11 | $I_{R1}$ and $V_{F1}$ | $I_{R1}$ and $V_{F1}$ | |
| 12 | See 4.3.1 | t = 48 hours (see 4.3.1) | |
| 13 2/ | Subgroups 2 and 3 of table I herein. $\Delta I_{R1}$ = 100 percent of initial value or 15 nA dc, whichever is greater. $\Delta V_{F1}$ = 30 mV dc. | Subgroup 2 of table I herein, $\Delta I_{R1}$ = 100 percent of initial reading or 30 nA whichever is greater; $\Delta V_{F1}$ = 55 mV dc. | |

1/ Thermal impedance shall be performed any time after sealing provided temperature cycling is performed in accordance with MIL-S-19500, screen 3 prior to this thermal test.
2/ Test within 24 hours after removal from test.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

High temperature reverse bias (HTRB), MIL-STD-750, method 1038, test condition A; $V_R$ = 12.8 V; $T_A$ = +150°C.

4.3.2 Screening (JANHC and JANKC). Screening of die shall be in accordance with MIL-S-19500, appendix H. As a minimum, die shall be 100-percent probed in accordance with group A, subgroup 2.



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4.3.3 Thermal impedance ($Z_{\theta JX}$ measurements). The $Z_{\theta JX}$ measurements shall be performed in accordance with MIL-STD-750, method 3101. The maximum limit (not to exceed the group A, subgroup 2 limit) for $Z_{\theta JX}$ in screening (table II of MIL-S-19500) shall be derived by each vendor by means of statistical process control. When the process has exhibited control and capability, the capability data shall be used to establish the fixed screening limit. In addition to screening, once a fixed limit has been established, monitor all future sealing lots using a random five piece sample from each lot to be plotted on the applicable X, R chart. If a lot exhibits an out of control condition, the entire lot shall be removed from the line and held for Engineering evaluation and disposition.

4.3.3.1 For initial qualification or requalification. Read and record data ($Z_{\theta JX}$) shall be supplied to the qualifying activity on one lot (random sample of 500 devices minimum). Twenty-two serialized devices shall be sent to the qualifying activity for test correlation.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein.

| | | |
|---|---|---|
| a. | $I_M$ measurement current | 1 mA to 10 mA. |
| b. | $I_H$ forward heating current | 50 mA to 200 mA. |
| c. | $t_H$ heating time | 10 ms. |
| d. | $t_{MD}$ measurement delay time | 70 µs maximum. |

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (for JANS) and table IVb (for JAN, JANTX and JANTXV) of MIL-S-19500, and as follows. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable inspections and footnotes of table I, group A, subgroup 2 herein.

4.4.2.1 Group B inspection, table IVa (JANS) of MIL-S-19500.

| Subgroup | Method | Condition |
|---|---|---|
| 4 | 1037 | $I_O$ = 33 mA; $V_R$ = 16 V(pk); f = 60 Hz; $T_A$ = room ambient as defined in the general requirements of 4.5 of MIL-STD-750; $t_{on}$ = $t_{off}$ = 3 minutes minimum for 2,000 cycles (see 4.5.1). |
| 5 | 1027 | $V_C$ = 16 V(pk); f = 60 Hz; $T_A$ = +125°C ±25°C for 240 hours with $I_O$ adjusted as required by the chosen $T_A$ to give an average lot $T_J$ = +150°C. Marking legibility requirements shall not apply (see 4.5.1). |
| 6 | 3101 or 4081 | $R_{\theta JL}$ = 250°C/W for type 1N5712-1 and $R_{\theta JL}$ = 700°C/W for type 1N5712; .375 inch (9.52 mm) lead length (nonsurface mount). $R_{\theta JEC}$ = 100°C/W (surface mount). |

4.4.2.2 Group B inspection, table IVb (JAN, JANTX and JANTXV) of MIL-S-19500.

| Subgroup | Method | Condition |
|---|---|---|
| 3 | 1027 | $I_O$ = 33 mA dc; $V_C$ = 16 V(pk); f = 60 Hz (see 4.5.1); $T_A$ = room ambient as defined in the general requirements of 4.5 of MIL-STD-750. |
| 4 & 5 | ---- | Not applicable. |
| 6 | 1032 | $T_A$ = +150°C. |

4.4.4 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500, and as follows. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable inspections and footnotes of table I, group A, subgroup 2 herein.



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| Subgroup | Method | Condition |
|---|---|---|
| 2 | 2036 | (Not applicable to surface mount devices); Lead fatigue conditions: Test condition E; .062 inch (1.57 mm) lead restriction from case. Tension conditions: Test condition A; 4 pounds, 15 seconds. |
| 3 | 2056 | Nonoperating. (Not required for double plug metallurgical bonded device) |
| 3 | 2006 | 20,000 g's; Y1, Y2, X1. (Not required for double plug metallurgical bonded device) |
| 5 | Not applicable | |
| 6 | 1026 | $I_O$ = 33 mA dc; $V_r$ = 16 V(pk); f = 60 Hz (see 4.5.1); $T_A$ = room ambient as defined in the general requirements of 4.5 of MIL-STD-750. |

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Steady-state operation life. This test shall be conducted with a half-sine wave of the specified peak voltage impressed across the diode in the reverse direction followed by a half-sine waveform of the specified average rectified current. The forward conduction angle of the rectified current shall not be greater than 180° nor less than 150°.

4.5.2 Thermal resistance. Thermal resistance measurement shall be in accordance with MIL-STD-750, method 3101 or 4081. Forced moving air or draft shall not be permitted across the device during test. The maximum limit for $R_{\theta JL}$ under these test conditions shall be as shown in group E. The following conditions shall apply when using method 3101:

a. $I_H$ - - - - - - - - - - - - - - - - - - - - - - - - - - - - - 1 mA to 10 mA
b. $I_H$ - - - - - - - - - - - - - - - - - - - - - - - - - - - - - 25 mA to 100 mA
c. $t_H$ - - - - - - - - - - - - - - - - - - - - - - - - - - - - - 25 seconds minimum
d. $t_{MD}$ - - - - - - - - - - - - - - - - - - - - - - - - - - - - 70 µs maximum

LS = Lead spacing = 9.53 mm (.375 inch) minimum for leaded devices and LS = 0 minimum for unleaded devices as defined on figure 4 below:

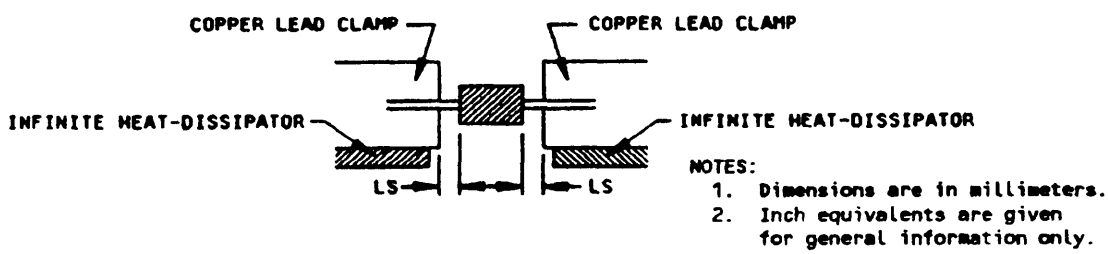



FIGURE 4. Mounting conditions.

4.5.3 Effective carrier lifetime ($\tau_{CL}$). This test shall be measured with the device installed in the test fixture, drawing C68001 using the test setup shown on figure 5 herein. Adjust the signal generator to 54 MHz and maximum output in order to obtain a sufficient trigger signal for the oscilloscope. Adjust the sensitivity of the oscilloscope to 20 mV/cm. With the device in the test fixture, adjust the output of the amplifier until the peak amplitude of the forward current is 20 mA or 5 cm as seen on the oscilloscope. Change sensitivity of scope to 2 mV/cm. Under these conditions, the effective carrier lifetime is related to the amplitude designated as "$\tau_{CL}$" shown on figure 6. This amplitude has the calibration of 50 ps/cm.



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TABLE I. Group A inspection.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 | | | | | | |
| Visual and mechanical inspection | 2071 | | | | | |
| Subgroup 2 | | | | | | |
| Forward voltage | 4011 | $I_F$ = 1 mA dc | $V_{F1}$ | --- | 0.55 | V dc |
| Forward voltage | 4011 | $I_F$ = 35 mA dc | $V_{F2}$ | --- | 1.0 | V dc |
| Reverse current | 4016 | DC method; $V_R$ = 16 V dc | $I_{R1}$ | --- | 150 | nA dc |
| Breakdown voltage | 4021 | $I_R$ = 10 µA dc | $V_{(BR)1}$ | 20 | --- | V dc |
| Thermal impedance | 3101 | See 4.4.1 | $Z_{\theta JX}$ | | | °C/W |
| 1N5712<br>1N5712-1, 1N5712UR-1 | | | | | 100<br>40 | |
| Subgroup 3 | | | | | | |
| High temperature operation: | | $T_A$ = +150°C | | | | |
| Reverse current | 4016 | DC method; $V_R$ = 16 V dc | $I_{R2}$ | --- | 150 | µA dc |
| Low temperature operation: | | $T_A$ = -55°C | | | | |
| Forward voltage | 4011 | $I_F$ = 1 mA dc | $V_{F3}$ | --- | 0.55 | V dc |
| Forward voltage | 4011 | $I_F$ = 35 mA dc | $V_{F4}$ | --- | 1.0 | V dc |
| Breakdown voltage | 4021 | $I_R$ = 10 µA dc | $V_{(BR)2}$ | 20 | --- | V dc |
| Subgroup 4 | | | | | | |
| Capacitance | 4001 | $V_R$ = 0 V dc; f = 1 MHz; $V_{sig}$ (pk) = 50 mV(pk) max | C | --- | | |
| 1N5712<br>1N5712-1, 1N5712UR-1 | | | | | 1.2<br>2.0 | pF<br>pF |
| Effective carrier lifetime (see 4.5.3) | --- | (see DESC drawing C68001) | $\tau_{CL}$ | --- | 100 | ps |
| Subgroups 5, 6, and 7 | | | | | | |
| Not applicable | | | | | | |

1/ For sampling plan, see MIL-S-19500.



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TABLE II. Group E inspections (all quality levels) for qualification only.

| Inspection 1/ | MIL-STD-750 | | Sampling |
|---|---|---|---|
| | Method | Conditions | |
| Subgroup 1 | | | 22 devices c = 0 |
| Thermal shock (temperature cycling) | 1051 | 500 cycles | |
| Electrical measurements | | See table I, group A, subgroup 2 ($V_{F1}$ and $I_{R1}$) | |
| Subgroup 2 | | | 22 devices c = 0 |
| Steady-state dc blocking life | 1038 | | |
| Electrical measurements | | See table I, group A, subgroup 2 ($V_{F1}$ and $I_{R1}$) | |
| Subgroup 3 | | | |
| Not applicable | | | |
| Subgroup 4 | | | 22 devices c = 0 |
| Thermal resistance surface mount | 3101 or 4081 | $R_{\theta JEC}$ = 100°C/W (maximum) at zero lead length. +25°C ≤ $T_R$ ≤ +35°C, (see 4.5.4) at $T_H$ ≥ 25 s. | |
| Thermal resistance leaded | 3101 or 4081 | For type 1N5712: $R_{\theta JL}$ = 700°C/W (maximum); For type 1N5712-1: $R_{\theta JL}$ = 250°C/W (maximum) at .375 inch lead length +25°C ≤ $T_R$ ≤ +35°C, (see 4.5.4) at $T_H$ ≥ 25 s in still air. | |
| Subgroup 5 | | | |
| Not applicable | | | |
| Subgroup 6 | | | 3 devices |
| ESD testing | 1020 | | |

1/ For sampling plan, see MIL-S-19500.



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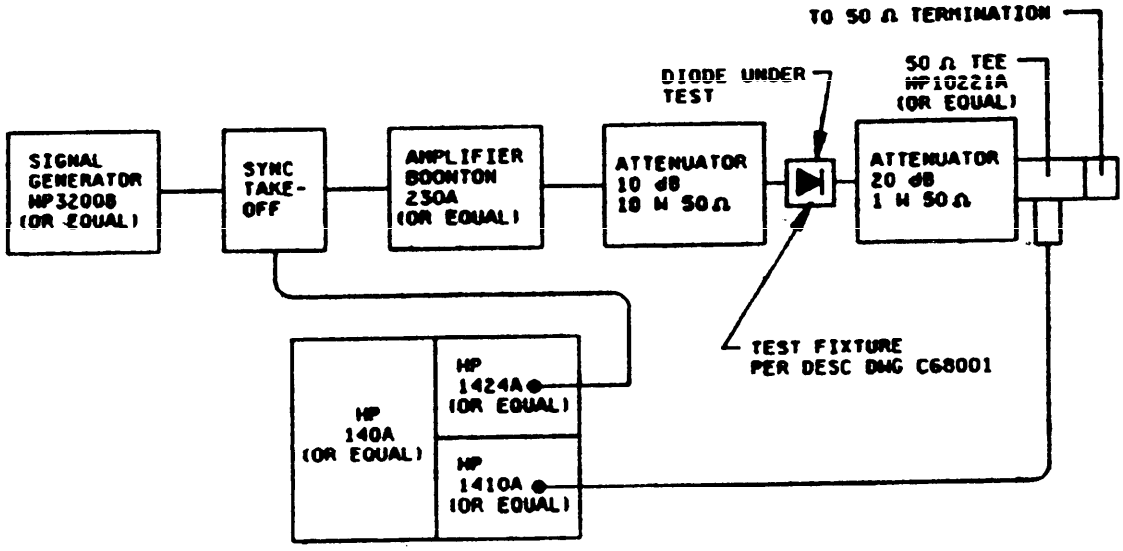



FIGURE 5. Test setup for diode lifetime measurement.

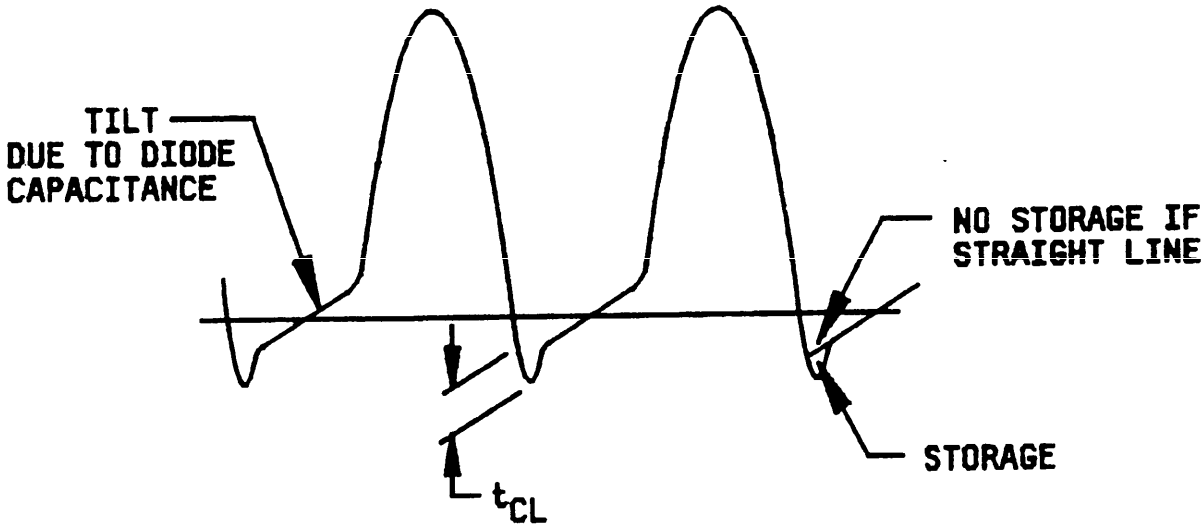



FIGURE 6. Oscilloscope display in diode lifetime measurement.



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4.5.4 Decap internal visual scribe and break. Scratch glass at cavity area with diamond scribe. Carefully snap open. Using 30X magnification examine the area where die (or bonding material) are in contact with the plugs, verify metallurgical bonding area.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead finish as specified (see 3.3.1).

c. Product assurance level, type designator, and for die acquisition, the JANHC and JANKC identification (see figure 3 and 6.3).

6.3 Suppliers of die. The qualified die suppliers with the applicable letter version (example JANHCA1N5712) will be identified on the QPL.

| JANC ordering information | | |
|---|---|---|
| | Manufacturer | |
| PIN | 55801 | -- |
| 1N5712 | JANHCA1N5712 | -- |
| 1N5712 | JANKCA1N5712 | -- |

6.4 Interchangeability information. The -1 device is not electrically identical to the non -1 device due to a difference in capacitance. Applications may be affected and users should evaluate their particularly applications for interchangeability.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC
Air Force - 19, 85, 99
DLA - ES

Preparing activity:
NASA - NA

Agent:
DLA - ES

(Project 5961-1569)



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